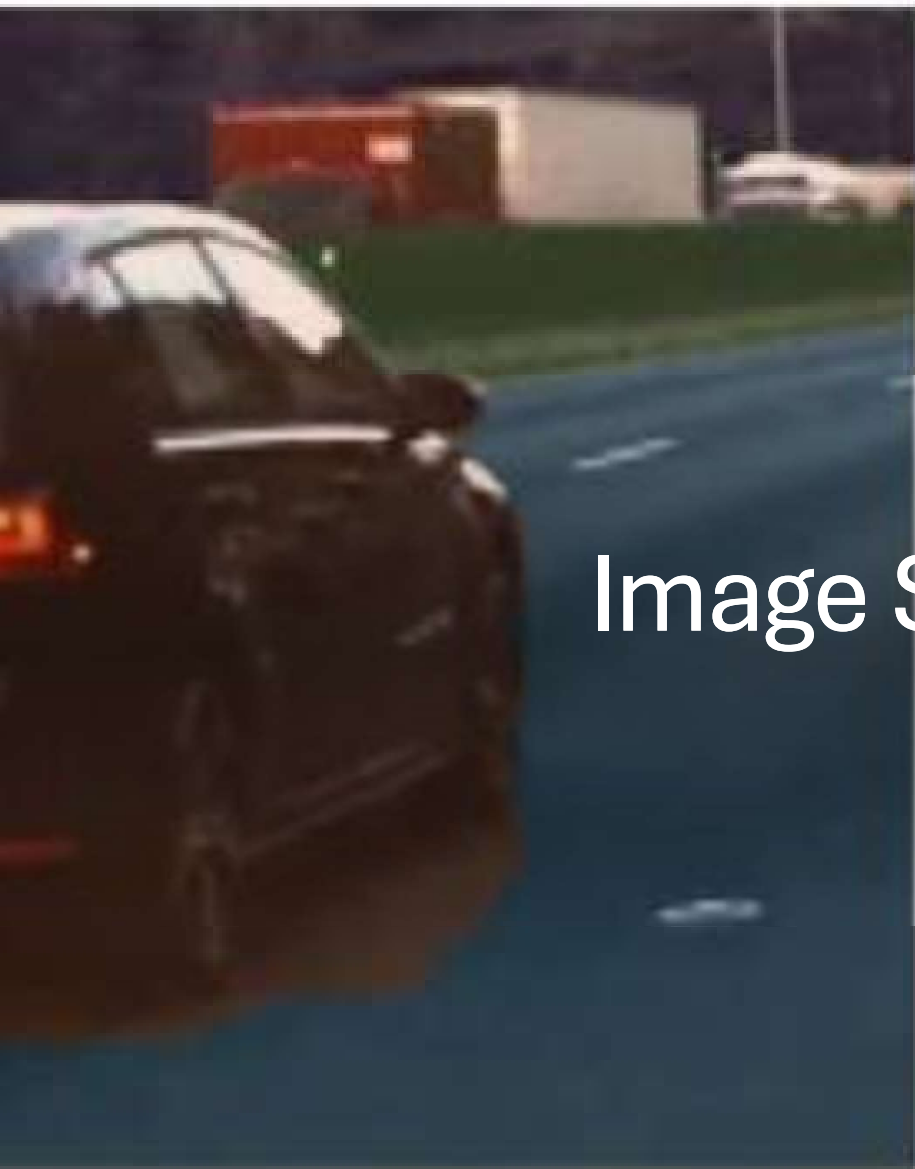
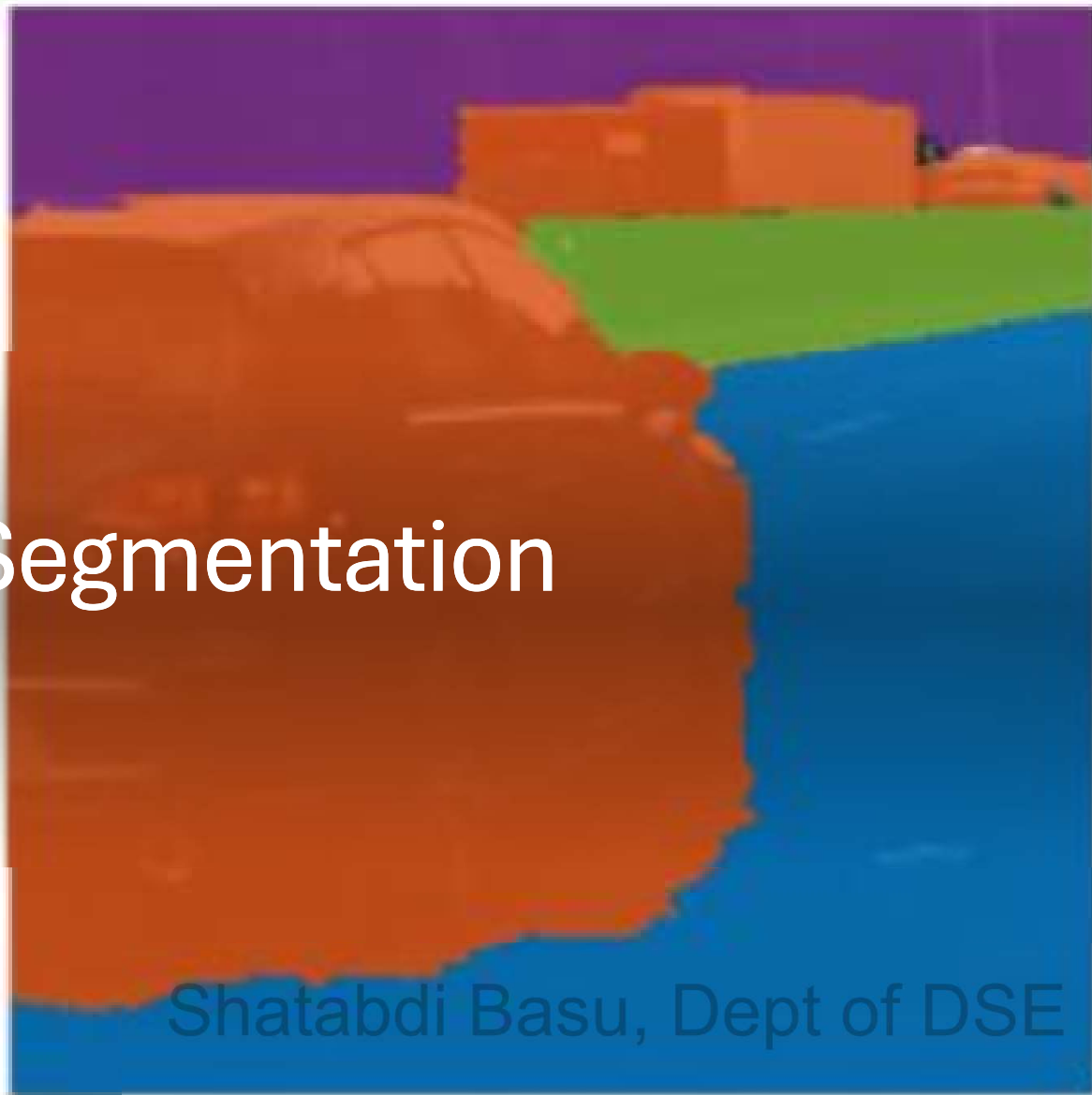




# Image Segmentation



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# Image segmentation

- The process of partitioning an image into multiple segments or regions to simplify or change the representation of the image.
- The goal is to identify and delineate objects, boundaries, and features within an image, allowing for more effective processing and analysis.



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# Aspects of image segmentation

- **Pixel Grouping:** Segmentation involves grouping pixels based on certain criteria, such as color, intensity, or texture, to form a region representing an object or a part of an object in the image.
- **Output:** The output of segmentation is often a set of regions or segments, each of which corresponds to different objects or parts within the image.
- **For example,** in a medical image, segmentation might be used to identify different anatomical structures.
- A segmentation image is all about assigning particular labels to pixels to identify objects, people, and other important elements.

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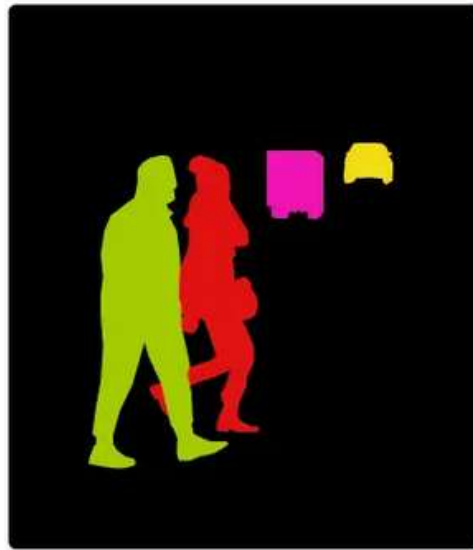
# Types of Image Segmentation

- **Semantic Segmentation** : This involves arranging the pixels in an image based on semantic class. **Example**: In an image of a street, all pixels that belong to "cars" will be labeled the same, even if there are multiple cars in the image.
- **Instance segmentation** : This technique involves classifying pixels based on the instances of an object instead of classes. **Example**: In an image with two cars, instance segmentation will assign different labels to each car, distinguishing them as separate entities.
- **Panoptic segmentation** : Combines both approaches, labeling each pixel with its semantic class and identifying individual object instances. **Example**: In an image of a street, each car will be treated as a separate instance (instance segmentation), while the sky, road, and buildings will be classified as background elements (semantic segmentation).

# Types of Image Segmentation



**SEMANTIC IMAGE  
SEGMENTATION**



**INSTANCE  
SEGMENTATION**



**PANOPTIC  
SEGMENTATION**

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# Applications of Image Segmentation

- **Object Recognition and Tracking :**
- By segmenting objects from the background, it becomes easier to extract relevant features and classify objects accurately.
- Object tracking algorithms utilize segmentation to track objects over time in video sequences.

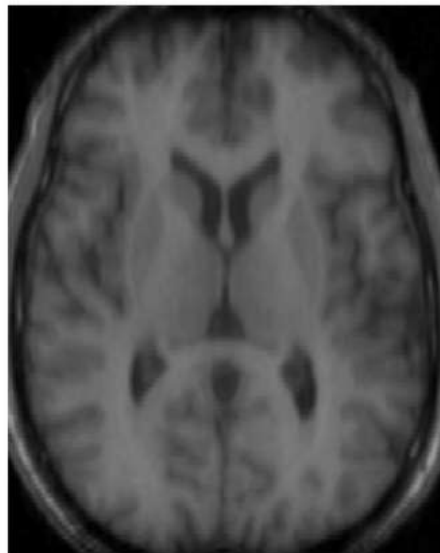


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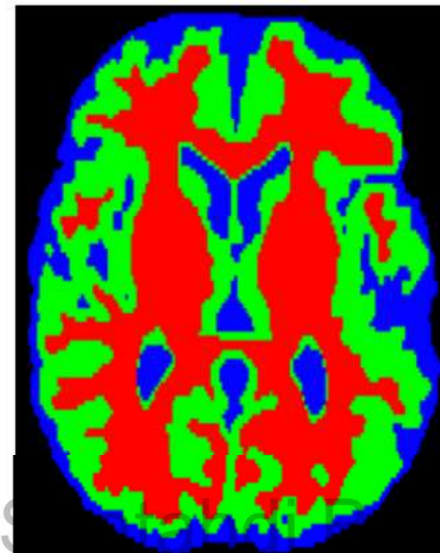
# Applications of Image Segmentation

- **Medical Imaging :**

- Image segmentation plays a critical role in medical imaging applications, such as tumor detection, organ segmentation, and disease diagnosis.
- Accurate segmentation of anatomical structures and abnormalities assists in surgical planning, treatment assessment, and computer-aided diagnosis.



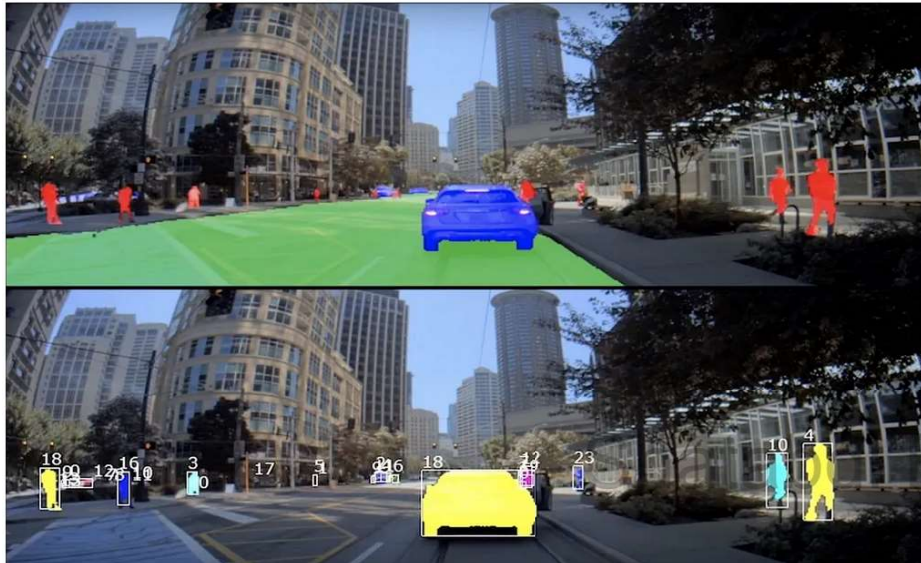
(a) Axial slice



(b) Tissue segmentation

# Applications of Image Segmentation

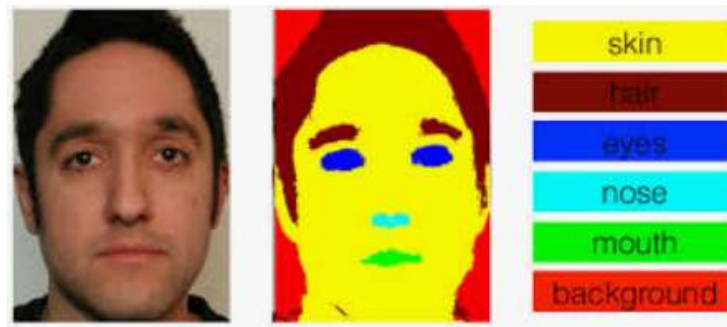
- **Autonomous Systems :**
- Segmentation is essential in autonomous systems, including autonomous vehicles and robots.
- It helps in scene understanding, obstacle detection, and navigation.
- By segmenting objects and the surrounding environment, autonomous systems can make informed decisions and navigate safely.





# Applications of Image Segmentation

- **Facial Recognition :**
- Image segmentation helps the AI-powered cameras identify unique features of each person's face that only a particular person can access on a phone or system.



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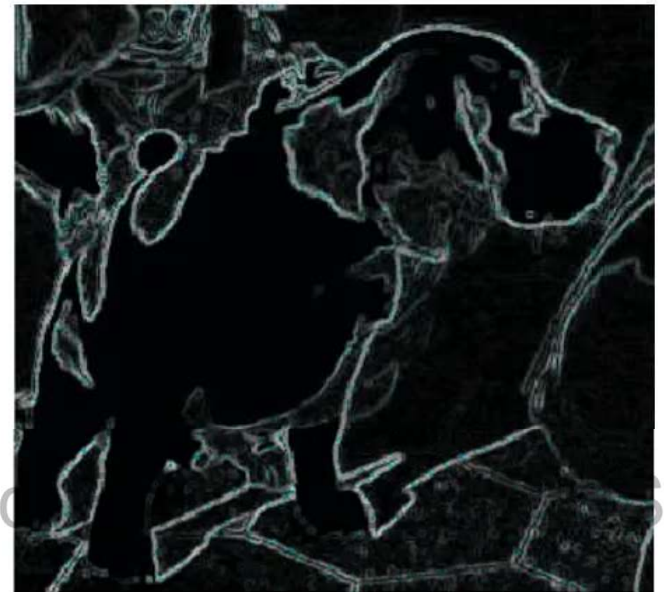
# Common Image Segmentation Techniques

- Edge-based segmentation
- Clustering-based segmentation
- Threshold-based segmentation
- Region-based segmentation

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# Edge-based Segmentation

- A technique used to partition an image by detecting the boundaries or edges between regions that exhibit significant changes in intensity or color.
- How edge-based segmentation works:
  - It begins by identifying points where intensity changes abruptly in the image (edges). Common edge detection operators are Sobel, Canny, Prewitt.
  - Once edges are detected, they are connected to form continuous boundaries around regions or objects.
  - Techniques like Hough Transform or active contour models (snakes) may be used to link and smooth the detected edges.



# Application of edge-based segmentation

- Detecting organ or tumor boundaries in X-rays, MRI, and CT scans.
- Identifying the contours of moving objects in video surveillance or self-driving cars.
- Extracting diagrams, shapes, or text boundaries from scanned documents.
- Detecting geographical features such as roads, rivers, and forest boundaries.
- Identifying defects on the surface of manufactured products (e.g., cracks or scratches).

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# Advantages and limitations of edge-based segmentation

- Advantages :
  - Works well when the boundaries between objects are well-defined.
  - Simple to implement and computationally efficient.
- Limitations :
  - Sensitive to **noise** in the image, which can cause false edges.
  - Struggles with images where **boundaries are not clearly defined** or have smooth transitions.
  - Requires **post-processing** (like edge linking) to obtain meaningful regions.

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# Clustering-based Segmentation :

- A technique that groups similar pixels into clusters based on their properties, such as intensity, color, or texture.
- The objective is to segment the image into meaningful regions, where each cluster corresponds to a region with similar characteristics.



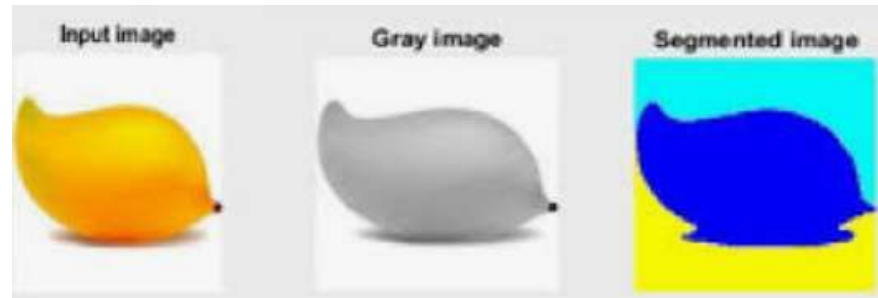
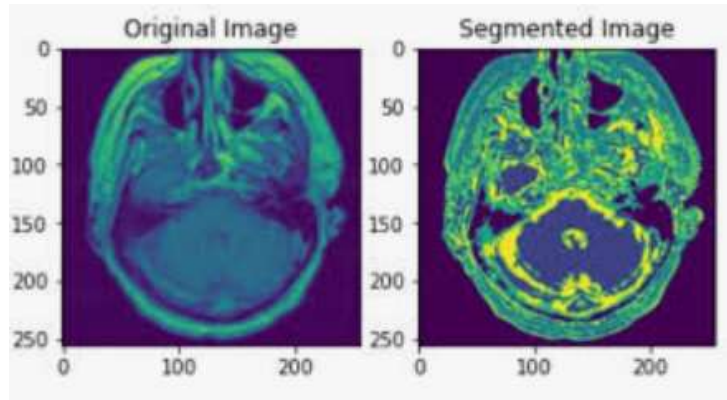
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# Common Clustering Algorithms for Image Segmentation

- K-Means Clustering
  - Divides the image into **K clusters**, where K is predefined.
  - Uses the **Euclidean distance** to assign pixels to the nearest cluster center.
- Fuzzy C-Means (FCM) Clustering
  - Each pixel can belong to multiple clusters with varying degrees of membership.
- Mean-Shift Clustering
  - Groups pixels based on **density estimation** (regions with high density are treated as clusters).
  - Does not require the number of clusters to be predefined.

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# Common Clustering Algorithms for Image Segmentation



**Fuzzy C-Means Clustering**

**K-Means Clustering**



**Mean Shift Clustering**

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# Application of clustering-based segmentation

- Identifying regions of interest like tumors or tissues from MRI and CT scans.
- Classifying regions (e.g., urban areas, forests, water bodies).
- Reducing the number of colors in an image by clustering pixels with similar colors.
- Detecting and segmenting similar objects or regions in an image (e.g., people in a crowd).

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# Advantages and limitations of clustering-based segmentation

- Advantages :
  - **Unsupervised:** Does not require labeled data.
  - **Effective for complex images:** Works well in scenarios with multiple regions and varying pixel intensities.
- Limitations :
  - **Choice of K:** In K-means, the number of clusters must be predefined.
  - **Sensitive to noise:** Performance can degrade with noisy or blurry images.
  - **Computational Complexity:** Some algorithms (e.g., Mean-Shift) can be computationally expensive.

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# Threshold-based Segmentation

- It involves setting a threshold value to separate pixels into different regions based on their intensity or color values.
- This technique is effective when the objects of interest have distinct intensities or colors compared to the background.
- Basic thresholding techniques include global and adaptive thresholding.
  - **Global thresholding** applies a fixed threshold value to the entire image. E.g- Otsu's method
  - **Adaptive thresholding** determines different threshold values for different image regions based on local characteristics. E.g- Adaptive mean thresholding, Niblack's and Sauvola's Thresholding



Original Image



Thresholded and segmented Image

# Application of threshold-based segmentation

- Segmenting tissues, organs, or tumors from medical scans like MRIs or X-rays.
- Identifying objects or regions of interest in surveillance videos or autonomous driving systems.
- Detecting defects such as cracks or scratches on product surfaces.
- Identifying faulty components during manufacturing.
- Separating text from the background in scanned documents.

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# Advantages and limitations of threshold-based segmentation

- Advantages :
  - **Simple and fast** to implement.
  - **Computationally efficient** for real-time applications.
  - Works well when the image has good **contrast** between objects and the background.
- Limitations :
  - **Sensitive to noise** and variations in lighting.
  - May not perform well on images with **smooth transitions** between objects and background.
  - Struggles with complex scenes where objects overlap or vary in intensity.

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# Threshold-based segmentation techniques

- Otsu's method
- Multivariable Thresholding
- Iterative Thresholding

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# Otsu's Method

- An **automatic global thresholding technique** that determines the optimal threshold value by maximizing the **between-class variance**.
- Working Methodology :
  - The image is divided into two classes: **foreground** and **background**.
  - Otsu's method finds the threshold that minimizes the **intra-class variance** and maximizes the **inter-class variance** (the variance between the two classes).
  - It assumes that the image has a **bimodal histogram** (two peaks representing foreground and background).

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# Multivariable Thresholding

- Multivariable thresholding, or **multi-thresholding**, refers to segmenting an image into **multiple regions** rather than just binary (foreground and background).
- Working Methodology :
  - Instead of finding a single threshold, the method finds **multiple threshold values** to partition the image into several regions.
  - This approach is useful when the image contains more than two meaningful areas (e.g., light, medium, and dark regions).

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# Iterative Thresholding

- A refined approach that gradually improves the threshold value through repeated iterations.
- Working Methodology :
  - The process starts with an **initial guess** for the threshold value.
  - In each iteration, the image is split based on the current threshold, and the **mean intensities** of the two regions (foreground and background) are computed.
  - A new threshold is calculated as the **average of the two means**, and the process repeats until the threshold stabilizes.

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# Difference between Thresholding Techniques

| Technique                  | Purpose                                         | Output                                          | Use Case                         |
|----------------------------|-------------------------------------------------|-------------------------------------------------|----------------------------------|
| Otsu's Method              | Finds optimal global threshold automatically    | Binary segmentation (foreground vs. background) | Medical imaging, OCR             |
| Multivariable Thresholding | Segments the image into multiple regions        | Multi-level segmentation (3+ regions)           | Satellite imagery, medical scans |
| Iterative Thresholding     | Refines threshold iteratively until convergence | Stable threshold through repeated iterations    | Noise-prone or complex images    |

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# Watershed Algorithm

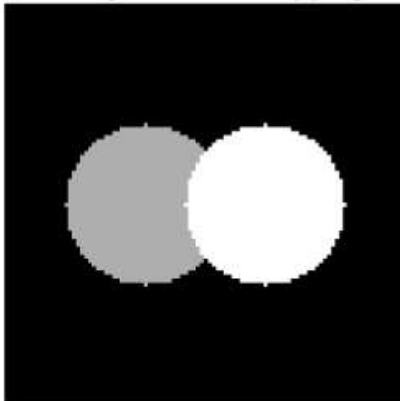
- A popular image segmentation technique inspired by the concept of watersheds in topography.
- It's often used to separate objects in an image based on their intensity values, especially when the objects are close together or overlapping.
- The basic idea is to treat a grayscale image as a topographic surface :
  - Dark areas represent "valleys" (low intensity).
  - Light areas represent "peaks" or "mountains" (high intensity).
- Water is imagined to start filling the valleys.
- As the water level rises, it will eventually start to overflow from one valley to another.
- At the points where water from different valleys would meet, boundaries or "dams" are placed.
- These boundaries become the segmentation lines that separate different regions in the image.

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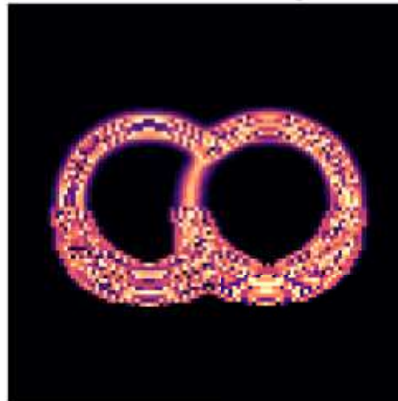
# Example of Watershed Algorithm

- Suppose we have a grayscale image of two overlapping circular objects.
  - **Compute the Gradient:** Detect the boundaries of the objects based on the intensity change in the image. This highlights the edges of each circle.
  - **Markers:** We label two markers, one in each circle (these can be chosen manually or automatically using connected components).
  - **Flooding and Dams:** Water starts to flood from each marker, filling up each circular region. As the water levels rise, they reach the edges of the circles where they overlap. Here, a dam or boundary is created, effectively separating the two circles.
  - **Result:** The output is a segmented image where each circle is distinctly separated, even though they were originally overlapping.

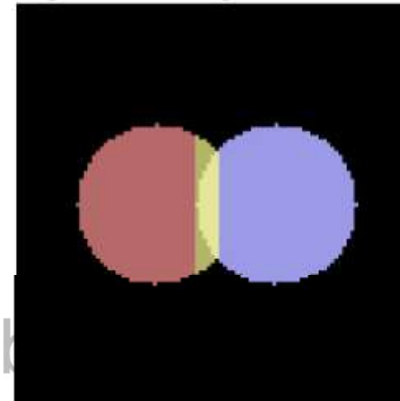
Original Image with Overlapping Circles



Gradient of Image



Segmented Image (Watershed)



take concept of DSE